higher education & training

Department:
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REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

(8080175)

1 August 2024 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 5 pages and a formula sheet of 5 pages.

110Q1G2401

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a black or blue pen.
5. Write neatly and legibly.

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QUESTION 1: ALTERNATING CURRENT THEORY

1.1 The RC integrator is obtained as soon as the output is taken from the capacitor.

Draw a neat labelled frequency response curve of an RC integrator. (2)

1.2 A coil with a 120 µH inductance and DC resistance of 10 Ω is connected in parallel with 150 nF capacitance. The combination is then connected to 1 kΩ in series. If the power supply is 10 V and 1 MHz, calculate the following values:

1.2.1 The total current (9)

1.2.2 The current through each branch (5)

[16]

QUESTION 2: POWER SUPPLIES

2.1 List TWO different types of filter circuits. (2)

2.2 A full-wave rectifier with a smoothing capacitor feeds a load of 1,0 kΩ. The AC frequency is 50 Hz and the average DC voltage is 20 V.

Calculate:

2.2.1 The value of the capacitor if the ripple voltage should not exceed 1 V peak to peak

2.2.2 The percentage ripple factor

(2 × 3) (6)

2.3 Use a labelled circuit diagram to briefly describe how a voltage tripler can be obtained. (7)

[15]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 The following values of a common emitter amplifier are known:

$Ic = 20\ mA$; $Re = 150\Omega$; $\beta = 200$ and $V_B = 1{,}7V$

Calculate the following values:

3.1.1 Vcc (4)

3.1.2 R_{B2} (2)

3.1.3 R_{B1} (2)

3.2 The following information on a common base amplifier that uses a PNP transistor is known:

R_C = 6,8 kΩ; R_L = 22 kΩ; R_E = 1,5 kΩ; R_C = 480 Ω;

h_{ib} = 40 Ω; $h_{rb} = 250 \times 10^{-6}$; h_{fb} = -0,96 and h_{ob} = 0,8 µA/V (Rs = 0)

Calculate the following values:

3.2.1 A_i (5)

3.2.2 A_v (2)

[15]

QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 List the properties of an operational amplifier. (3)

4.2 Draw a neat, labelled circuit diagram of a differential amplifier using two transistors. (5)

4.3 A differential amplifier which uses NPN-transistor, uses a ± 16 V supply voltage. Re has a value of 2 kΩ and the testing collector voltage, Vce, is +10 V. The current gain, *hfe*, is 50 and the input impedance, *hie*, is 800 Ω.

Calculate the following values:

4.3.1 R_L (4)

4.3.2 Voltage gain (2)

[14]

QUESTION 5: INTEGRATED CIRCUITS

Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (5.1–5.4) in the ANSWER BOOK.

5.1 To test the static voltage that a human body invokes as soon as a person runs across a polished floor.

5.2 The charges are destructive as soon as they charge on the inputs of C-MOS circuits.

5.3 The diodes are connected in a forward-biased configuration between the input (gate) and positive supply.

5.4 Voltage pulses that exceed ± 15 V, are chopped.

(4 × 1) **[4]**

QUESTION 6: ELECTRONIC PHASE CONTROL

Draw a neat, labelled circuit diagram that can be used for speed control of low-power series motors and universal motors. **[6]**

QUESTION 7: TEST EQUIPMENT

7.1 With the aid of a block diagram, show a frequency counter that counts the number of pulses of cycle signals per time unit. (5)

7.2 Explain the function of a Schmitt trigger circuit in a frequency counter. (1)
[6]

QUESTION 8: OSCILLATORS

8.1 Oscillators are unstable amplifiers that use positive feedback.

Give FOUR types of oscillators using positive feedback. (4)

8.2 Draw a neat, labelled circuit diagram of an A-stable multivibrator. (4)

8.3 A Schmitt trigger is one of the components used to convert signals.

State the operation principle of a Schmitt trigger circuit. (4)
[12]

QUESTION 9: TRANSDUCERS

9.1 9.1.1 Draw a neat, labelled circuit diagram of a closed-loop system in which a synchronous motor is used to control a radar antenna. (5)

9.1.2 Explain what happens as soon as a transmitter rotor moves through an angle of θ. (2)

9.2 9.2.1 Draw a neat, labelled circuit diagram of a relay circuit controlled by a LASCR. (3)

9.2.2 List TWO functions where LASCRs can also be used. (2)
[12]

TOTAL: 100

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FORMULA SHEET

$$I = \frac{V}{R} \qquad P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + ... = I_1R_1 + I_2R_2 + I_3R_3 + ...$$

$$I_T = I_1 + I_2 + I_3 + ... = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + ...$$

$$T = RC \qquad T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt} \qquad V_C = \frac{1}{RC}\int v_i dt$$

$$X_L = 2\pi fL \qquad X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L \qquad Z = R - jX_C$$

$$Z = R + j(X_L - X_C) \qquad I_T = \frac{V_T}{Z_T}$$

$$V_R = I_T R \qquad V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C) \qquad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$BW = f_2 - f_1 \qquad \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2} \qquad I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{R(jX_L)}{R + jX_L} \qquad \frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L \qquad I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C} \qquad \frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C \qquad I_T = \frac{V}{R} + j\frac{V}{X_C}$$

$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

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$I_T = I_R - j(I_L - I_C)$

$I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$

$a + jb = \sqrt{a^2 + b^2} \underline{/\tan^{-1}\frac{b}{a}} = r \underline{/\theta}$

$r\underline{/\theta} = r(\cos\theta + j\sin\theta)$

$Q = \tan\theta$

$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$

$Z_d = \frac{L}{CR_1}$

$V_{rms} = \frac{1}{\sqrt{2}} V_m = 0{,}707\, V_m$

$V_{dc} = \frac{2}{\pi} V_m = 0{,}637\, V_m$

$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P}$

$V_{dc} = \frac{1}{\pi} V_m = 0{,}318\, V_m$

$PIV = V_m$

$PIV = 2\, V_m$

$R_{r(rms)} = 0{,}385\, V_m$

$V_{r(rms)} = 0{,}305\, V_m$

$r = \frac{V_{r(rms)}}{V_{dc}}$

$V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$

$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$

$V_{dc} = V_m - \frac{I_{dc}}{2fC}$

$V_{dc} = V_m - \frac{I_{dc}}{4fC}$

$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}fC} = \frac{V_{dc}}{2\sqrt{3}fCR_L}$

$V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{V_{dc}}{4\sqrt{3}fCR_L}$

$r = \frac{I_{dc}}{2\sqrt{3}fCV_{dc}} = \frac{1}{2\sqrt{3}fCR_L}$

$r = \frac{I_{dc}}{4\sqrt{3}fCV_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$

$V'_{dc} = \frac{R_L}{R + R_L} . V_{dc}$

$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} . V_{r(rms)}$

$X_C = \frac{1}{2\pi fC}$ $X_C = \frac{1}{4\pi fC}$ $r' = \frac{V'_{r(rms)}}{V'_{dc}}$

$V'_{r(rms)} = \frac{X_C}{R} . V_{r(rms)}$

$r' = rX_C\left(\frac{R + R_L}{R.R_L}\right)$

$V'_{dc} = V_{dc} - I_{dc}R_1$

$V'_{dc} = \frac{R_L}{R_L + R_1} . V_{dc}$

$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC}$

$V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$

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$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$2V_m = V_{c2} = V_m + V_{c1}$$

$$3V_m = V_{c1} + V_{c3} = V_m + 2V_m$$

$$S = \frac{\Delta V_o}{\Delta V_i}$$

$$V_R = V_i - V_z$$

$$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}}$$

$$I_z = \frac{P_z}{V_z}$$

$$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} . R_S$$

$$V_o = V_r - V_{be}$$

$$R_c = \frac{V_{cc} - V_{ce}}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be}}{I_b}$$

$$\beta = \frac{I_c}{I_b}$$

$$C_e \geq \frac{10}{2\pi f R_e}$$

$$V_e = \frac{V_{cc}}{10}$$

$$R_e = \frac{V_e}{I_e} \simeq \frac{V_e}{I_c}$$

$$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$$

$$R_{b2} = \frac{1}{10} \beta R_e$$

$$V_b = V_e + V_{be}$$

$$V_{be} = h_{ie} i_b + h_{re} V_{ce}$$

$$i_c = i_{fe} i_b + h_{oe} V_{ce}$$

$$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L}$$

$$A_i = h_{fe}$$

$$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L} \right) \left(\frac{R_b T}{R_{bT} + Z_1} \right) \left(\frac{R_c}{R_c + R_L} \right)$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L}$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie}}$$

$$Z_1 = h_{ie} - \frac{h_{fe} h_{re} Z_L}{1 + h_{oe} Z_L}$$

$$Z_1 = h_{ie}$$

$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_s}}$$

$$Z_2 = \frac{1}{h_{oe}}$$

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$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i$ $A_p = \frac{h_{fe}{}^2 R_L}{h_{ie}}$

$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$

$Z_0 = R_C // Z_2 = Z_L // Z_2$ $Z_1 = R_b // Z_1$

$Z_i = R_{b1} // R_{b2} // Z_1$ $I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$

$I_0 = h_{fe} I_b = h_{fe} \left(\frac{R_{b2}(I_i)}{R_{b2} + h_{ie}} \right)$ $A_i = \frac{I_0}{I_1}$

For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.

$Z_L = R_c // R_L$ $I_1 = \frac{R_e I_i}{R_e + Z_1}$

$CMRR = \frac{A_{dm}}{A_{cm}}$ $CMRR\ (dB) = 20log \frac{A_{dm}}{A_{cm}}$

$I_e = \frac{V_e}{R_e}$ $I_c = \frac{I_e}{2}$

$R_L = \frac{V_{R_L}}{I_C}$ $g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$

$V_0 = -\left(\frac{R_f}{R_1} \right) . V_i$ $V_0 = \left(\frac{R_f}{R_1} + 1 \right) . V_i$

$V_0 = -\left(\frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3 \right)$ $V_0 = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) R_f$

$V_0 = -(V_1 + V_2 + V_3)$ $V_0 = -(I_1 + I_2 + I_3)\ R_f$

$V_0\ (t) = -\frac{1}{RC} \int V_i(t)$ $V_0\ (t_b) = -\frac{1}{RC} \int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$

$t = \frac{1}{f}$ $A_v = -\frac{R_s}{R_1}$

$R_2 = \frac{R_1 R_s}{R_1 + R_s}$ $f_c = \frac{1}{2\pi R_s C}$

$V_0(t) = -RC \frac{dV_i(t)}{dt}$ $A = \frac{R_f}{X_c}$

$A = -\frac{R_f}{R_s}$ $V_0(t) = -R_f C \frac{d}{dt} . v_i \sin \omega t$ \

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$$t = R_f C \qquad V_0 = A(V_r - V_i)$$

$$V_0 = \frac{R_f}{R_s}(V_2 - V_1) \qquad V_0 = V_2 - V_1$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}} \qquad f_0 = \frac{1}{2\pi RC}$$

$$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}} \qquad L_T = L_1 + L_2 + 2M$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_T}} \qquad C_T = \frac{C_1 C_2}{C_1 + C_2}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_2}} \qquad f = \frac{1}{2\pi RC\sqrt{6}}$$

$$f_0 = \frac{1{,}5}{RC} \qquad f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$$

$$t_1 = 0{,}7\ R_2 C_1 \qquad t_2 = 0{,}7\ R_1 C_2$$

$$f_0 = \frac{1}{1{,}4RC} \qquad V_i = I_{c2} R_e + V_{be(ON)}$$

$$t = 1{,}1\ RC \qquad f_0 = \frac{1{,}443}{(R_A + 2R_B)C}$$

$$t_{low} = 0{,}693\,(R_B)C \qquad t_{high} = 0{,}693\,(R_A + R_B)C$$

$$t_T = t_{low} + t_{high} \qquad K = \frac{\Delta R / R}{\Delta l / l}$$

$$\sigma = \Delta l / l \qquad R = \rho \frac{1}{\pi d^2 / 4}$$

$$\sigma = \frac{S}{E} \qquad Resolution = \frac{1}{number\ of\ turns}$$

$$Resolution = \frac{voltage\ drop\ across\ adjacent\ turns}{total\ voltage\ drop}$$

$$R_t = Ae^{B/T} \qquad T = 273 + {}^\circ C$$

$$V_A = \frac{R_2}{R_1 + R_2}.V_T \qquad V_B = \frac{R_t}{R_t + R_3}.V_T$$

$$V_{AB} = V_A - V_B \qquad A_v = \frac{V_0}{V_i}$$

$$V_{Hall} = kIH$$

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